

KARIM AYMAN ABD-ELMAQSOOD KHALIL

Structural BIM Technician (Modeler) | Technical Office Draftsman

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PROFESSIONAL SUMMARY

Structural BIM Technician (Modeler) and Technical Office Draftsman with 5+ years of experience in structural drafting, BIM modeling, reinforcement detailing, and construction documentation across nuclear power plant and wastewater treatment projects. Skilled in building structural Revit models from scratch and developing consultant-provided models progressing from LOD 350 to LOD 400 according to project requirements, with LOD 500 where specifically required, including model audits, warnings management, drawing generation, and quantity takeoff. Produced 300+ As-Built drawings and 120+ Bar Bending Schedules (BBS), plus structural, reinforcement, waterproofing, execution, and repair-stage documentation from Detailed Design Drawings (DDD). Applies GOST, Rosatom, and NPPA requirements and coordinates with Egyptian and Russian engineering teams to resolve technical RFIs, drawing revisions, and site-driven changes.

CORE SKILLS

BIM & CAD Software: Autodesk Revit (Structural), AutoCAD, Civil 3D, Autodesk Construction Cloud (ACC), Dynamo (Basic)

Geospatial Tools: ArcGIS

Model Development: 3D Structural Modeling from Scratch, Consultant Model Development (LOD 350 to LOD 400; LOD 500 where specifically required)

Model Coordination: Multidisciplinary Coordination, Clash Identification & Resolution, Model Updates, Workset Management, Naming Conventions

Drawing Production & QTO: Drawing Generation, Quantity Takeoff (QTO), Sheet Management, Drawing Scales, Title Blocks

Templates & Standards: BIM Standards Implementation, Family Management, View Templates, Sheet Standards

Quality & Mentoring: Model Audits, Warnings Management, Model Health, Quality Checks, Mentoring

Technical Office & Detailing: Reinforcement Detailing, Bar Bending Schedules (BBS), Sections & Elevations, Execution Drawings, Waterproofing Systems, Construction Joints, Bay Boundaries, Repair Stages

Documentation: As-Built Drawings, Technical RFI Resolution, Drawing Revisions, Submission Packages

Standards & Frameworks: Detailed Design Drawings (DDD), Work Execution Plans (WEP), BIM Execution Plans (BEP), Exchange Information Requirements (EIR), GOST, Rosatom, NPPA Nuclear Project Requirements

PROFESSIONAL EXPERIENCE

Structural BIM Technician (Modeler) | Technical Office Draftsman

Feb 2023 – Present

Hassan Allam Construction (HAC) | El Dabaa Nuclear Power Plant, Units 3 & 4 | El Dabaa, Egypt — Client: NPPA | Main Contractor: ASE / Rosatom

- Developed 4+ structural 3D Revit models from scratch and from consultant-provided models, progressing from LOD 350 to LOD 400 per project requirements.
- Produced 300+ As-Built drawings covering reinforcement, waterproofing systems, and repair stages, integrating verified site data into final documentation.
- Created structural, reinforcement, waterproofing, and execution shop drawings from Detailed Design Drawings (DDD) in line with GOST, Rosatom, and NPPA requirements.
- Produced 120+ Bar Bending Schedules (BBS) for execution-ready structural elements.
- Applied Work Execution Plans (WEP) to bay boundaries, construction joints, and waterproofing interfaces across 30+ concrete bays.
- Coordinated with Egyptian and Russian engineering teams to resolve 150+ site RFIs and implement approved technical changes while maintaining design intent across revisions.
- Performed Revit model audits and warnings management to maintain model health, consistency, and quality across structural packages.
- Maintained drawing scales, sheet numbering, title blocks, and view templates across shop-drawing sets per documentation requirements.
- Guided and reviewed the work of junior draftsmen, providing technical feedback on drafting methods and project standards.
- Prepared and organized drawing submission packages for client and consultant review, ensuring completeness and revision control.

Technical Office Draftsman

Mar 2022 – Feb 2023

MEGA Construction | El Dabaa Nuclear Power Plant, Unit 1 | El Dabaa, Egypt — Client: NPPA | Main Contractor: ASE / Rosatom

- Developed structural shop drawings and complex reinforcement configurations directly from DDD packages for nuclear construction works.
- Prepared execution drawings for concrete bays per the Work Execution Plan (WEP), including sensor locations, feed points, and construction joints.
- Reviewed and revised reinforcement shop drawings to resolve complex clashes and convert approved technical solutions into execution-ready layouts.
- Produced structural sections, elevations, and construction layouts in AutoCAD, incorporating red-line markups and site RFI changes while maintaining drawing standards and version control.
- Coordinated with the Russian engineering team and surveying engineers to prepare final As-Built drawings from verified field measurements.

Technical Office Draftsman

May 2021 – Mar 2022

Orascom Construction PLC | Abu Rewash Wastewater Treatment Plant | Giza, Egypt — Client: CAPW | Main Contractor: Orascom Construction PLC & Aqualia

- Drafted P&ID diagrams for water treatment processes from mechanical engineering inputs.
- Produced reinforced-concrete structural drawings and shop drawings for water tanks and plant facilities.
- Developed mechanical equipment layouts and piping route drawings with interdisciplinary coordination.
- Prepared final As-Built documentation for piping and structural elements.

CERTIFICATIONS & TRAINING

- Autodesk Revit 01: Structure Modeling — Hassan Allam Holding, Sep 2026
- Revit Architecture Essentials — Autodesk Authorized Training Center, Jun 2026
- Revit Structure Essentials — Autodesk Authorized Training Center, Apr 2026
- Revit Quantity Takeoff — Autodesk Authorized Training Center, Jan 2026
- Introduction to BIM — Autodesk Authorized Training Center, Dec 2025
- AutoCAD — Advanced Drafting & Shop Drawings, 2020

EDUCATION

Bachelor of Arts in Geography — Geographic Information Systems (GIS), Cartography & Surveying

Tanta University | Graduated 2021

LANGUAGES

Arabic: Native

English: Professional Working Proficiency (technical reading, writing, and documentation)